

Charles Darwin University
PRT661 Data Science Practice Unit

Assessment 1

Project Proposal and Design

Solar Generation Forecasting for Battery Dispatch

DKASC Alice Springs · 2008–2026

Theme 2: Predictive Analytics and Forecasting

Group 4

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Repository: github.com/Uttam1618/PRT661-solar-forecasting

Jira board: [SCRUM — PRT661 Solar Forecasting Group 4](#)

Submitted 16 August 2026

1. Project overview and objectives

Group 4 will predict photovoltaic generation at the Desert Knowledge Australia Solar Centre (DKASC) site in Alice Springs, 1 to 24 hours in advance, to inform a site owner or operator on when a battery should charge or discharge. Forecast and uncertainty are the deliverable, it's not a system that we build, but the scheduling of the battery is the operational use it informs. Four goals are identified: create a batch pipeline that can be reproduced for nineteen yearly extracts (2008-2026) containing some 1.9 million rows and 3.1 GB of data; develop degradation-aware and ramp-sensitive features; evaluate proposed models against a published benchmark for this site; and display forecasts and forecast errors in a dashboard. It has a design that covers all five of the capabilities: acquisition, storage, processing, analytics/machine learning and visualisation.

2. Theme justification

The project is located in Theme 2, Predictive Analytics and Forecasting, which predicts a continuous future outcome based on historical data, with time-series forecasting being explicitly mentioned in the examples in the theme. Theme 1 was rejected, since dispatch decision does not need to be done with the minimum latency, so it is a batch pipeline and not a streaming one. The reason for excluding Theme 3 is because per-inverter anomaly channels are out of scope, and the fifty or so per-array columns are not included in the feature set for this reason.

3. Problem statement

Solar generation at Alice Springs is dependent on the presence of cloud and on the season, plus a slow degradation of the solar panels as they age. A battery only pays for itself if the operator can predict what generation will be available, and this battery energy storage system was installed in 2024 at the site. Thuseethan et al. (2025) predict this same location, but accuracy is an objective, they use 2019 to 2024, and they pre-date the battery. Three gaps follow. The 2024 installation provides dispatchable SoC (State of Charge) signals. The 17-year period shows degradation not apparent in a 5-year period. The previous studies reported high error at dawn/dusk and peak irradiance but did not investigate what aspects cause the high error. This project is designed to forecast to support operational decision and directly tackles ramp events and feature importance. We are going to use their best published result - R^2 0.8641 and MSE 22.41 on Master Meter 2 to measure up against.

4. Initial architecture

PRT661 Solar Generation Forecasting High-Level System Architecture

DKASC Alice Springs · 2008–2026 · Theme 2 Predictive Analytics and Forecasting

Objective: forecast generation 1–24 hours ahead so that battery charge and discharge can be scheduled against expected output

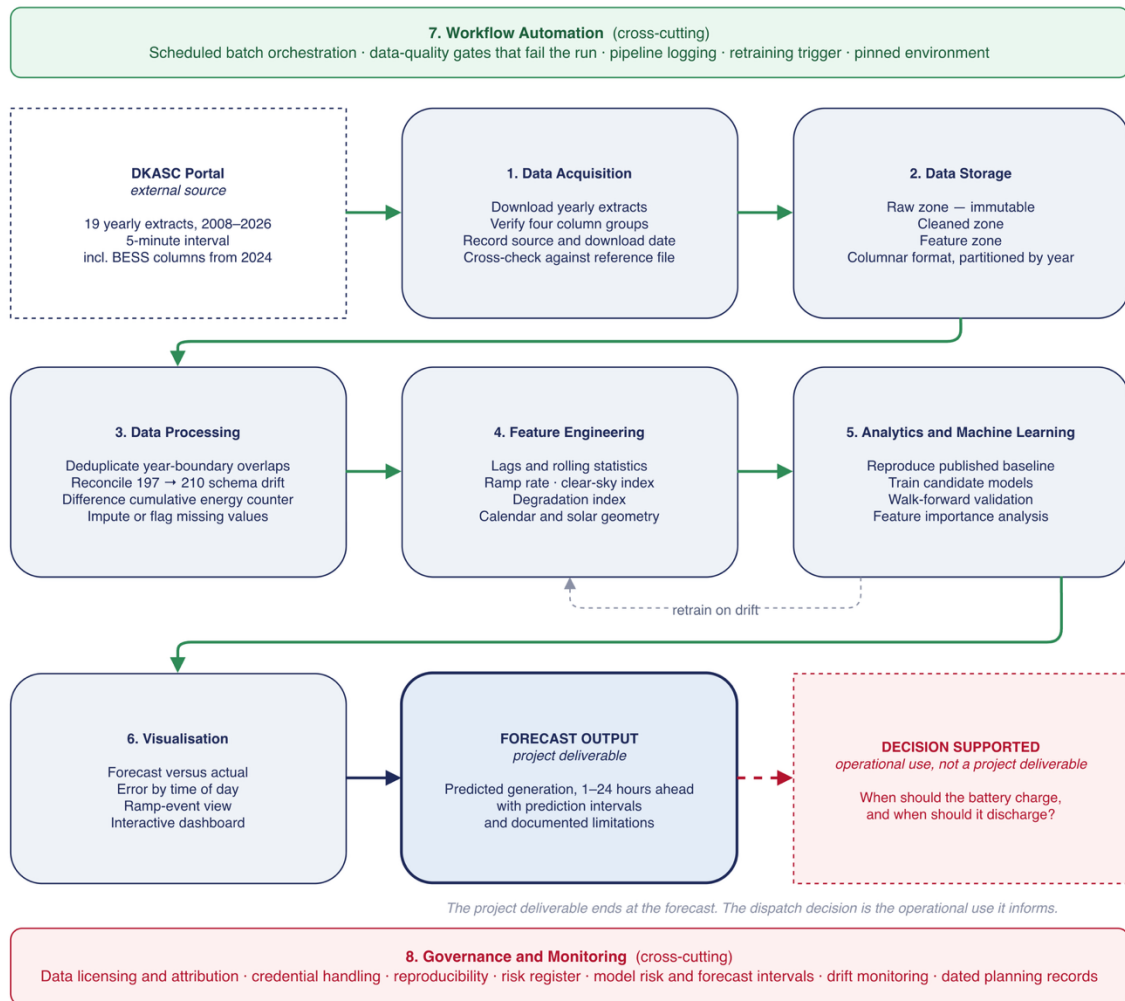


Figure 1. High-level system architecture. The forecast output is computed independently of the dispatch decision that it provides.

The high-level design is shown in Figure 1. The data follows six stages from the DKASC portal to a forecast, with workflow automation and governance spanning all six stages. The forecast is distinct from the decision, which is the edge of what this project will provide. The storage uses parquet files queried through DuckDB; PostgreSQL was rejected as at 1.9 million rows, the overhead would not be justified, and Apache Spark as the working set would be in memory. These both add to the cost of operation, but do not enhance the results. All components are open source and can be reproduced from the repository and Figure 2 shows the design, layer-by-layer.

Data Pipeline Architecture — Layered View

Batch pipeline · scheduled execution · all open-source components



Figure 2. Data pipeline architecture layered view: component, output format and rebuild cadence at each layer.

5. Workflow plan

Workflow Plan — Epics, Owners and Milestones

Every epic name below appears identically as a Jira epic in the project board



Figure 3. Workflow plan: the eight epics against the four assessment milestones.

The following eight epics are shown against the four assessment milestones in figure 3. The eight names of epic in figure 3 are the eight epic words on the Jira board, and they are worded exactly the same, and each epic is assigned to its accountable owner, as a Jira assignee. The board has 18 tasks,

each of which has been assigned to a member of the team and to a sprint, seven of which are Done, four In Progress and seven To Do, across four sprints corresponding to the A1, A2, A3 and A4 work areas. Progress, scope changes and completion status are recorded as work takes place and not regenerated when it is submitted.

6. Task allocation and team structure

Group 4 has four members. Table 1 provides a list of roles and epic ownership: each member owns two epics across the entire pipeline order with the Project Lead also having named tasks within Feature Engineering and Analytics and Machine Learning as second member, ensuring that each epic has two members who are familiar with it. Work is done when it is committed, reviewed by another member and their Jira task closed with an evidence link. Decisions are made documented on dated planning records and not agreed verbally.

Role	Member	Epics (accountable)	Report sections
Project Lead	Uttam Shrestha	Workflow Automation, Governance and Monitoring	5, 6, 9 and assembly
Data Engineer	Abhishek Tamang	Data Acquisition, Data Storage	1, 3
Data and Feature Engineer	Sarin Uprety	Data Processing, Feature Engineering	4, 7
Modelling and Visualisation Lead	Yogesh Basnet	Analytics and Machine Learning, Visualisation	2, 8

Table 1. Roles, epic ownership and report sections.

7. Initial risk analysis

The repository's risk register lists 19 risks under the categories of technical, project and governance, including likelihood, impact, mitigation and a risk owner. The seven most significant, by impact, are shown in Table 2. The register is checked at each stand up meeting and each submission; review of 12 August resulted in giving owners to named members and thus matching each risk to a named Jira assignee. Just two risks have been realised and mitigated; the first being the slippage against the internal draft deadline and the second being a stale extract which ends 11 February 2026.

ID	Risk	Impact	Mitigation
T1	Schema drift: 197 columns to 2023, 210 from 2024	Medium	The pipeline takes care to reconcile both schemas explicitly, indicating that the BESS columns are only available from 2024 onwards.
T3	Cumulative counter is giving the wrong readings as interval energy.	High	The cumulative series is differenced, checked for validity with the reference extract and unit-tested prior to use.
T4	High missingness: wind speed 52%, tilted radiation ~35%	Medium	Wind speed is removed and the other missing values are imputed or flagged, and it is noted which of these was done.

ID	Risk	Impact	Mitigation
T5	Stale extract: 2026 data ending on 11th February 2026	Medium	Before A2, the data is re-downloaded and the download date is noted; and A1 is truthful about its coverage.
T8	Overfitting through inappropriate validation	High	No random shuffling at any time, instead it is walking forward validation.
P1	Uneven contribution across the group	High	All work has a named Jira assignee, is discussed weekly in the stand-up and issues are escalated upfront rather than on submission.
G4	Poor dispatch decision due to inaccurate forecast.	Medium	Prediction intervals are reported for forecasts and known weak periods are openly reported as limitations.

Table 2. Extract from the initial risk register, rated by impact.

8. Ethics, privacy and security

No personal information is included in the dataset. It simply documents electrical data from a demonstration facility and data from a co-located weather station, and no one can be identified in either case, thus no consent process is necessary. Data here and in repository is credited to DKASC, and source URL and date of download recorded to allow results to be traced back to a specific extract; raw files are not version controlled.

Seasonal calendars of First Nations were used in previous work on this site. We intentionally do not emulate that approach. This is because it is the substantive work of the authors, and Indigenous seasonal knowledge is cultural intellectual property and should only be used following engagement with Traditional Owners, which is beyond the scope and timeframe of this unit.

There is an operational cost to a poor forecast – not just a statistical cost. Prediction intervals are thus reported along with point forecasts, known weak time periods are recorded, and the system is stated to be a decision support system, not an automated controller. Credentials are never committed: .gitignore excludes environment, key and credential files. Where generative AI may be used to aid in drafting or coding, its use is identified in accordance with CDU policy.

9. References

- Data download | DKA solar centre. (n.d.). In *dkasolarcentre.com.au*. Retrieved 16 August 2026, from <https://dkasolarcentre.com.au/download?location=alice-springs>
- Thuseethan, S., Gangajaliya, S., Hamlin, L., Shanmugam, B., & Thennadil, S. (2025). Conv-ensemble for solar power prediction with first nations seasonal information. *IEEE Open Journal of the Computer Society*, 6, 884–895. <https://doi.org/10.1109/OJCS.2025.3580339>

Project Update Summary

Changes Since Previous Assessment

Area	Previous State (A1)	Current State (A2)	Justification
Primary dataset	19 yearly extracts, 2008 to 2026	Master Meter 1 extract; yearly files retained for schema evidence and 2024-onward telemetry	The yearly extracts carry no weather columns at all. Master Meter 1 does, and is the benchmarked series.
Dataset scale	~1.9 M rows, 3.1 GB	1,824,033 rows measured (3.11 GB); Master Meter 1 holds 1,767,903	A1 carried an estimate; both are now measured.
Schema drift	197 columns to 2023, 210 from 2024	406 distinct names; 14 in every year, 3 of those empty; 13 new at 2024	Measured across all 19 files. The 2024 additions are BESS, PV, grid and site-demand telemetry.
Cumulative counter	Difference and unit test (risk T3)	Six digit register that wraps at 1,000,000; 6 wraps corrected, corrupt reads nulled	Differencing alone gave minus one million per wrap. Repaired totals agree with power-integrated energy to 99.7%.
Degradation	Degradation-aware features proposed	-0.454 kW per year (-0.94% of mean), fitted from 2014	Fitting from 2009 returns +0.748 kW per year because the site was still being commissioned.
Model specification	Models to be evaluated against a benchmark	Gradient boosting at 1 h and 24 h; three baselines; MAE, RMSE, R2, MBE; 80% quantile intervals; 4-fold walk forward	Closes the A1 feedback that these were unspecified.
Jira evidence	Board named, no URL given	Direct board URL supplied, staff access enabled	A1 feedback: claimed tasks and assignees could not be verified.

Contribution Summary

Team Member	Contribution
Uttam Shrestha	PLACEHOLDER - acquisition, storage, automation, governance, report assembly
Abhishek Tamang	PLACEHOLDER - data processing
Sarin Uprety	PLACEHOLDER - feature engineering
Yogesh Basnet	PLACEHOLDER - analytics, ML, visualisation

Evidence Locations

Evidence	Repository Location
Schema profile and drift	notebooks/01, 01b; Outputs/schema_matrix.csv, schema_drift_summary.md
Ingestion and quality baseline	notebooks/02; Outputs/data_quality_mastermeter1.md
Cleaning, before and after	notebooks/03; Outputs/cleaning_report_mastermeter1.md
Features	notebooks/04; Supporting_Documents/feature-dictionary.md
Exploratory figures	notebooks/04b; Outputs/figures/fig1 to fig6
Models, metrics, intervals	notebooks/05; Outputs/model_results.md; figures fig7 to fig10
Risk, ethics, governance, PM	Supporting_Documents/; Planning/; Task_Allocation/; Project_Planning_Records/

Assessment 2 Technical Development

1. Scope, and what changed since Assessment 1

Assessment 1 proposed a batch pipeline over nineteen yearly DKASC extracts, degradation-aware and ramp-sensitive features, evaluation against a published benchmark for the same site, and a dashboard. The operational purpose was unchanged: forecast photovoltaic generation one to twenty four hours ahead so that a site operator can decide when the battery should charge, hold or discharge.

Profiling the archive before writing any ingestion code changed the design in one decisive way. The nineteen yearly extracts contain 1,824,033 rows and 406 distinct column names, but **not one meteorological column**. They record per-inverter electrical readings and nothing that explains them. Building a solar forecast on that archive would mean modelling generation without irradiance.

The Master Meter 1 reference extract, downloaded from the same portal, contains 1,767,903 rows at the same five minute resolution and carries global horizontal radiation, diffuse horizontal radiation, tilted radiation, air temperature, relative humidity, wind speed and direction, and daily rainfall alongside the target. It is also the exact series on which Thuseethan et al. (2025) report their results, so it makes benchmark comparison possible rather than approximate. Master Meter 1 therefore became the primary dataset. The yearly extracts are retained as schema evidence and as the source of the site-level battery telemetry available from 2024.

This is a documented change of direction, not a reduction in scope. The pipeline, the feature families, the benchmark and the operational framing are all as proposed. Every figure and table below is reproducible from the notebooks named in the Evidence Locations table.

2. Updated architecture

The layered design in Assessment 1 survives contact with the data. What changed is the acquisition layer, which now reads one curated extract rather than reconciling nineteen drifting schemas, and the addition of an explicit profiling stage ahead of ingestion.

Stage	Implementation	Output	Rebuild cadence
Profiling	notebooks/01, 01b. Header and schema scan across all 19 extracts	schema_matrix.csv, schema_drift_summary.md	On each new download
Acquisition	notebooks/02. Master Meter 1 extract, timestamp integrity enforced against a complete 5 minute grid	mastermeter1_5min.parquet	On each new download
Storage	Snappy-compressed Parquet, columnar, queried through DuckDB	5 minute and hourly stores	Rebuilt from raw
Processing	notebooks/03. Sentinel removal, physical ranges, register repair, bounded gap interpolation	mastermeter1_5min_clean.parquet, hourly_clean.parquet	Rebuilt from raw
Feature engineering	notebooks/04. Solar geometry, clear sky index, calendar, lags, rolling statistics, degradation	features_hourly.parquet	Rebuilt from clean
Analytics and ML	notebooks/05. Gradient boosting, three baselines, walk forward validation, quantile intervals	model_results.md, figures 7 to 10	Per evaluation run
Visualisation	notebooks/04b, 05. Ten figures covering data quality, exploration and model behaviour	Outputs/figures/	Per run
Governance	Cross-cutting. Quality gates, risk register, evidence links	Supporting_Documents/	Continuous

Two Assessment 1 technology decisions were revisited against measured figures and both stand. PostgreSQL remains unjustified: the working set is 148,537 hourly rows and fits comfortably in memory, so a server process would add operational cost without improving any result. Apache Spark remains unjustified for the same reason. Parquet with DuckDB gives columnar reads, so the modelling notebook loads only the columns it needs from a 31 MB feature store rather than re-parsing a 390 MB CSV.

The separation Assessment 1 insisted on is preserved throughout: **the system produces a forecast and an interval, not a dispatch instruction**. Section 9 sets out how an operator turns one into the other, which is a human decision supported by the output.

3. The five Vs of this dataset

Assessment 1 addressed the five Vs in prose. They are tabulated here with measured values, as recommended in the Assessment 1 feedback.

V	Measured position	Consequence for the design
Volume	3.11 GB across 19 yearly extracts (1,824,033 rows) plus a 390 MB reference extract (1,767,903 rows). Working set after hourly aggregation: 148,537 rows, 31 MB.	Large enough to rule out spreadsheet tooling and to require columnar storage; small enough that a distributed engine would be pure overhead.
Velocity	Batch. The portal publishes periodically and the current extract ends 11 February 2026 for the yearly files and 23 August 2025 for Master Meter 1. No streaming source.	Confirms Theme 2 rather than Theme 1. The pipeline rebuilds on download rather than ingesting continuously; dispatch decisions are hourly, not sub-second.
Variety	406 distinct column names across the archive, of which only 14 appear in every year and 3 of those are empty. 13 columns appear at 2024. Electrical, meteorological and battery telemetry in one archive.	Schema reconciliation is a first-class pipeline stage, not an afterthought. It also drove the choice of Master Meter 1, the only extract with a stable schema and weather.
Veracity	Sentinel codes (999999, -99999), physically impossible values (irradiance to 2,725 W/m ² , humidity to 131%, temperature to -40 C), a six digit register that wraps, and missingness from 2.3% on the target to 52.8% on wind speed.	Every cleaning rule is derived from instrument physics rather than the data's own percentiles, and the register repair is validated against an independent estimate.
Value	A 1 hour forecast with MAE 3.61 kW and an 80% interval, and a 24 hour forecast with MAE 6.96 kW, against a site whose hourly output ranges to 209 kW.	Sufficient to inform battery charge and discharge scheduling at both the intra-day and day-ahead horizons, with quantified uncertainty at each.

4. Data acquisition and storage

Profiling ran first, deliberately, so that ingestion was written against what the files contain rather than what Assessment 1 assumed. Row counts were taken without parsing the files into memory, and only the header and three sample rows of each extract were read to build the column matrix.

The result corrected two claims carried in Assessment 1. The archive holds **1,824,033 rows, not approximately 1.9 million**, and the schema situation is far more severe than the reported step from 197 to 210 columns. Across nineteen files there are 406 distinct column names. Only fourteen appear in every year, and three of those (the archived M15 QCells channels) are entirely empty in recent years, leaving eleven columns with unbroken coverage: the timestamp and the M1 array across its three phases. The remaining 392 columns are inverters commissioned and decommissioned across seventeen years of site operation.

The thirteen columns appearing at 2024 were confirmed as the site-level telemetry added with the battery installation: BESS active, reactive and apparent power plus state of charge, and PV, grid and site demand power totals. This is a stronger position than Assessment 1 anticipated, because surplus (photovoltaic generation less site demand) and battery headroom (state of charge) are both directly measured from 2024 onward. They are reserved for Assessment 3, where the dispatch layer is built.

A quieter finding matters for anyone reproducing this work: **each yearly file overruns one day into the following year**. The 2015 extract, for instance, covers 1 January 2015 through 1 January 2016 at 23:55, exactly 366 days at 288 readings per day. Naively concatenating the nineteen files would silently duplicate roughly 288 rows at each of eighteen boundaries. Using the single Master Meter 1 extract avoids the problem entirely, and the finding is recorded against SCRUM-16.

Ingestion enforces timestamp integrity by reindexing onto a complete five minute grid spanning the observed period. This converts implicit gaps, where a reading simply does not appear, into explicit missing values. Without it, missingness is invisible and any lag feature silently spans an unknown interval. The reindex inserted 14,519 empty slots into a 1,782,422 slot grid, and found zero duplicate timestamps and zero unparseable timestamps.

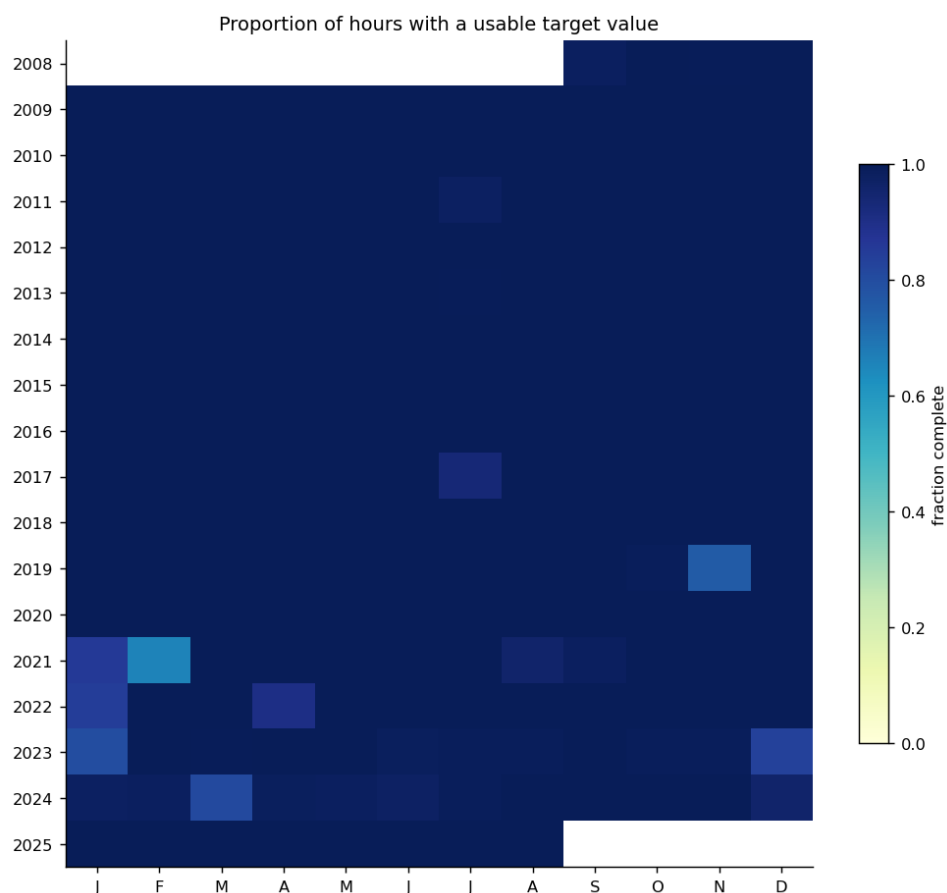


Figure 4. Proportion of hours carrying a usable target value, by month. Overall completeness is 99.06%. The weakest months are February 2021 at 65.8% and November 2019 at 75.7%. White bands at the start of 2008 and the end of 2025 are the limits of the extract, not outages.

Storage is Snappy-compressed Parquet at both five minute and hourly resolution, queried through DuckDB. Raw and derived data are excluded from version control by `.gitignore`; the small analysis artefacts under `Outputs/` are explicitly re-included so that every number in this report is traceable to a committed file.

5. Cleaning and preprocessing

Cleaning is applied at five minute resolution and the hourly store is rebuilt afterwards. The ordering is deliberate and was corrected during development: an initial version aggregated first, which allowed sentinel values to propagate into hourly means and left a minimum air temperature of -39.988 C in the hourly store.

Five stages are applied in sequence.

Stage	Rule	Rationale	Effect
Sentinel removal	Absolute value at or above 9,999 in any <i>instrument</i> channel becomes null	Loggers write fixed error codes rather than blanks. No instrument here has a legitimate reading of that magnitude.	Removes -99,999 wind directions and tilted radiation values in the tens of thousands
Register diagnosis	The cumulative energy register is exempt from the rule above and diagnosed separately	A cumulative kWh counter is not an instrument reading. At a mean load near 45 kW over seventeen years it legitimately accumulates millions of kWh; applying an instrument bound to it destroys the column.	Identified a six digit register that wraps at 1,000,000, with 6 wraps

Stage	Rule	Rationale	Effect
Register repair	Wraps have 1,000,000 added back; remaining negatives and jumps above 1,000 kWh per 5 minutes become null	Clipping negative differences to zero would fabricate intervals of no generation.	Validated: 6,467,319 kWh from the register against 6,488,854 kWh integrated from Active Power, agreement 99.7%
Physical ranges	Per column bounds from instrument limits and the Alice Springs climate record	Percentile-based outlier detection lets a systematic fault define what counts as normal. Recorded local temperature extremes are roughly -7.5 C and 45.2 C.	Humidity capped at 100%, tilted diffuse radiation from 97,322 to 767 W/m2
Bounded interpolation	Time weighted, for runs of at most 3 consecutive slots (15 minutes). The target is never interpolated.	A 5 minute gap in irradiance is safely bridged; a six hour gap is not. Filling the variable being predicted manufactures agreement between model and data.	281,355 values interpolated; the target remains at its measured 2.34% missing

Hourly aggregation then requires at least six of the twelve possible readings in an hour, otherwise the hour is nulled. An hourly mean over two surviving readings is not comparable to one over twelve, and treating them alike would quietly corrupt the training set. The resulting store holds 148,537 hourly rows of which 147,134 carry a usable target.

One interpretation deserves stating precisely, because a naive reading of the data would get it wrong. **Roughly half of all Active Power readings are negative**, and this is not an error. The meter is signed: the site draws a small standby load overnight, so night-time values sit near -0.5 kW while daytime values rise to 209 kW. Likewise, the energy register records net energy delivered less received, so small negative steps overnight are real consumption rather than corruption. Approximately 743,880 such steps occur, and describing them as corrupt would misrepresent 43% of the record.

Cleaning cannot repair absence. Wind speed remains 52.8% missing, power factor, current phase average and total harmonic distortion each around 41.7%, and the two tilted radiation channels 34% to 37%. These were never recorded rather than recorded wrongly. They are excluded from the feature set on completeness grounds, which confirms the exclusion of wind speed anticipated in Assessment 1 as risk T4.

6. Feature engineering

Thirty features are built in five families. The governing constraint is that no value at time t may use information recorded at or after t : every lag and rolling statistic is backward shifted, and the contemporaneous target is never an input to its own prediction. This is what makes the walk forward results in Section 8 meaningful rather than self-confirming.

Family	Features	Why
Solar geometry	solar_elevation, ghi_clearsky, is_daylight	Elevation from declination, the equation of time and the hour angle; clear sky irradiance from the Haurwitz model. Physics rather than learned parameters, so these cost no training data and extrapolate correctly to unseen years.
Cloud	clearsky_index and its lags and 24 hour rolling mean	Measured irradiance divided by modelled clear sky irradiance. Removes the daily and seasonal cycle and leaves the cloud signal, which is the genuinely unpredictable part. Undefined at night by construction, which the model handles natively.
Calendar	hour_sin, hour_cos, doy_sin, doy_cos	Cyclical encodings so that hour 23 sits adjacent to hour 0 rather than 23 units away.

Family	Features	Why
Lag and rolling	target_lag{1,2,3,24,168}, ghi_lag{1,2,24}, csi_lag{1,2,24}, target_roll{3,24,168}, csi_roll24, target_std24	Lag choice is justified by the partial autocorrelation in Figure 5 rather than asserted. target_std24 carries recent volatility, which is what makes an interval widen on unsettled days.
Degradation	years_since_start	Captures panel ageing across seventeen years, quantified in Figure 1.

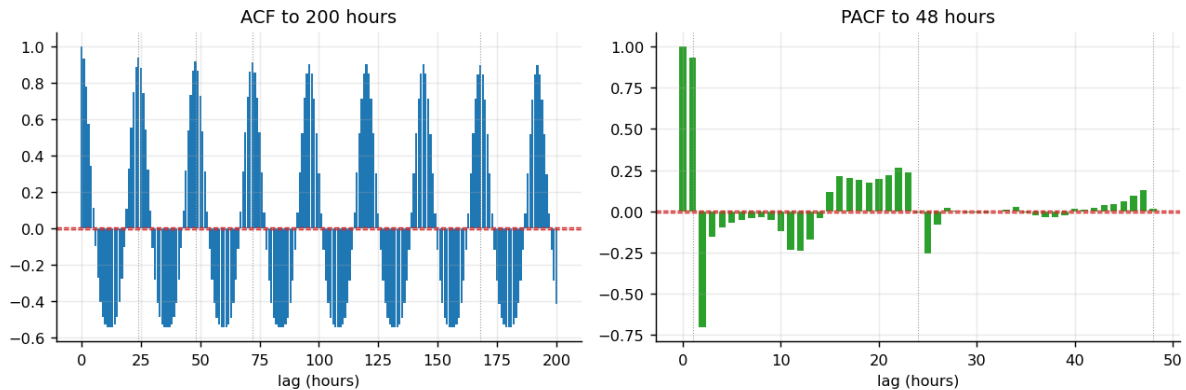


Figure 5. Autocorrelation and partial autocorrelation of hourly Active Power, computed on the complete hourly grid with a time aware shift so that lag k is exactly k hours. The ACF peaks at multiples of 24 confirm that the daily cycle dominates and justify the 24 and 168 hour lags; the PACF identifies which shorter lags carry information the others do not already contain.

An earlier version of this analysis correlated by position within a daylight-only series. With roughly twelve daylight hours per day, position 24 in that series is about two days, so the axis labelled hours was not hours. Excluding night is correct when cross correlating with features, where a shared overnight zero inflates the relationship, and wrong for autocorrelation, where the daily cycle is the signal. The distinction is worth recording because the incorrect version looked entirely plausible.

7. Exploratory analysis

Six figures were produced, each intended to defend a claim this report makes rather than to decorate it. Three carry findings that changed the analysis.

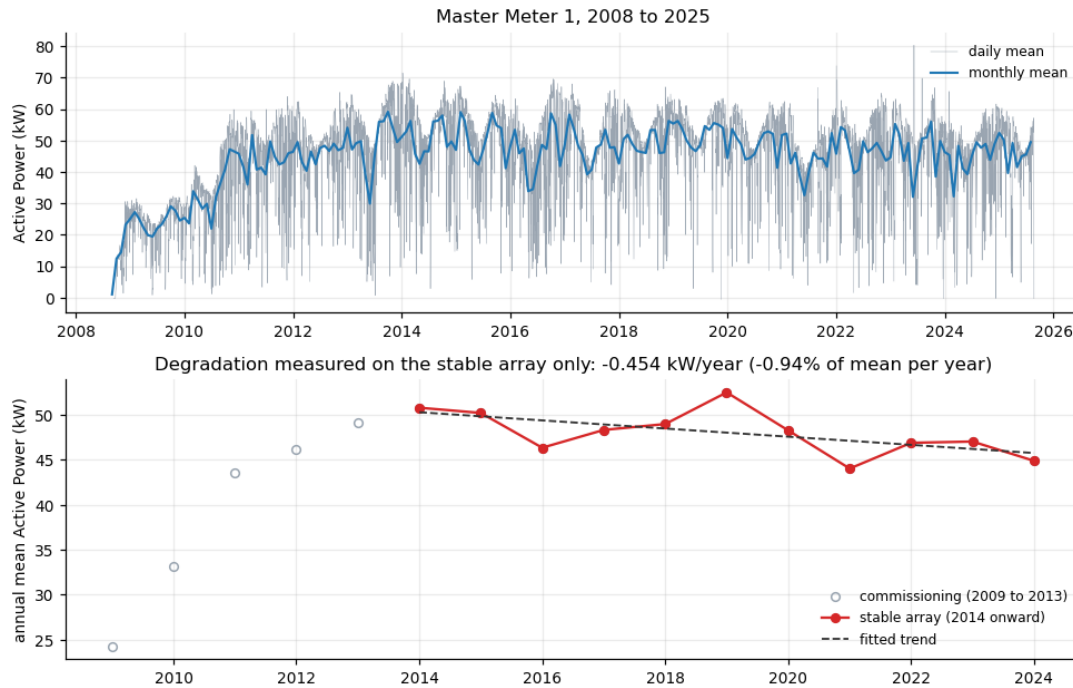


Figure 1. Seventeen years of Master Meter 1 output. Upper: daily mean (grey) and monthly mean (blue). Lower: annual means, with the 2009 to 2013 commissioning period shown as open markers and the trend fitted only from 2014.

Degradation is measured at -0.454 kW per year, or -0.94% of the mean annually. The windowing matters more than the number. Fitting across 2009 to 2024 returns +0.748 kW per year, because the site was still being commissioned in the early years and output rises as arrays are added rather than falling as panels age. A degradation chart showing generation increasing would have been an obvious error in a report that claims degradation-aware features. Restricting the fit to the stable array gives a figure inside the 0.5% to 1% band typically published for crystalline silicon, recovered from seventeen years of site data with no tuning.

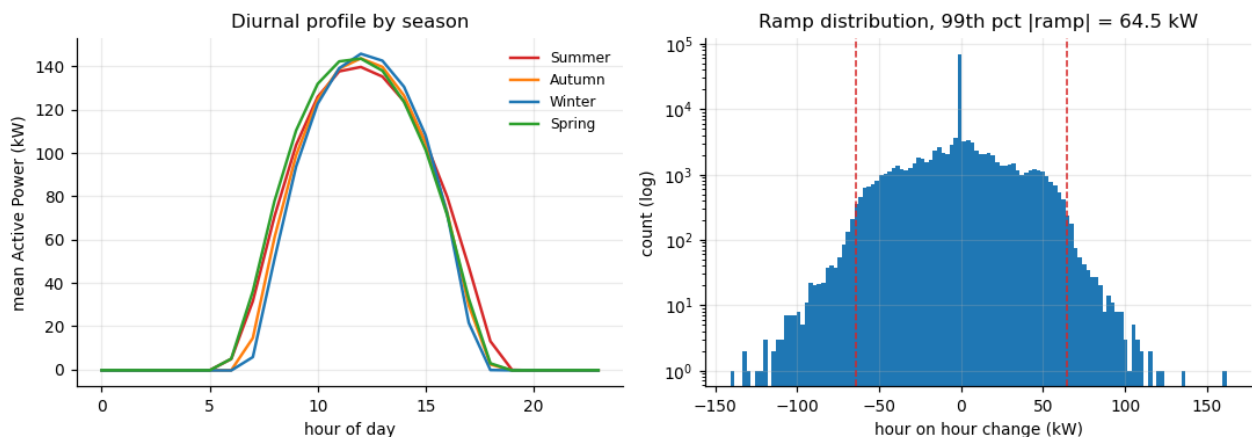


Figure 2. Left: mean diurnal profile by season. Right: distribution of hour on hour change, log scaled, with the 99th percentile of absolute ramp at 64.5 kW marked.

Two features of the diurnal panel look wrong and are not. Winter peaks marginally above summer, at roughly 143 kW against 140 kW, because Alice Springs sits almost on the Tropic of Capricorn so the winter noon sun is still high, winter skies are clearer, and cooler panels are more efficient. Summer is wider rather than taller: the seasonal energy difference comes from day length, not peak intensity. That has a direct operational consequence, since a battery has more hours in which to absorb surplus in summer.

The ramp panel quantifies the problem Assessment 1 set out to address. There are 1,470 hours whose change exceeds 64.5 kW, with extremes of +162.1 kW and -140.8 kW. Against a site peaking near 209 kW, a single hour can swing more than two thirds of full output.

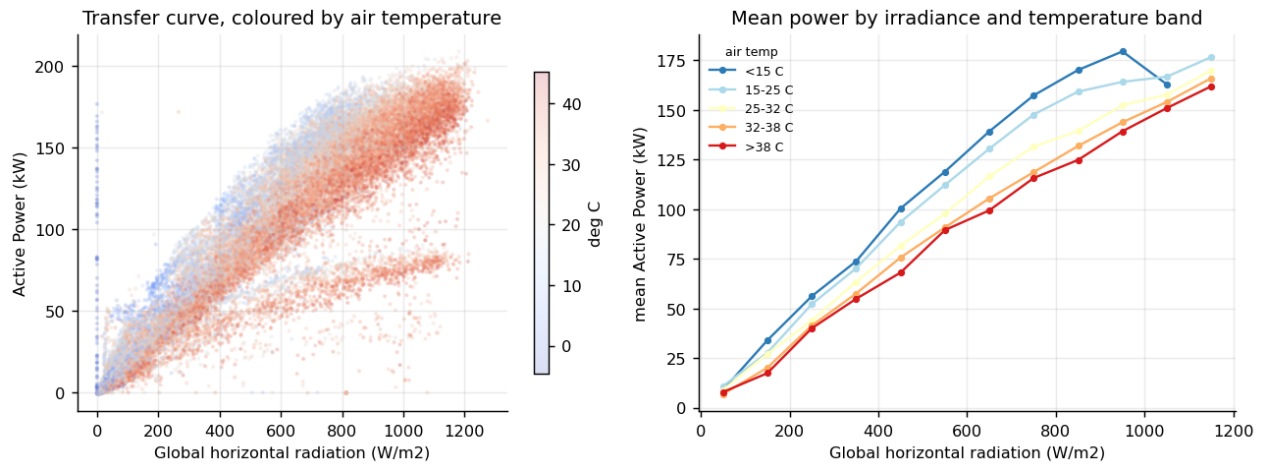


Figure 6. Left: the physical transfer curve, coloured by air temperature. Right: mean power by irradiance band, split by temperature band.

Irradiance and power correlate at $r = 0.903$ across daylight hours. The right panel shows the temperature effect: at 700 to 800 W/m², mean output falls from 157.3 kW below 15 C to 115.7 kW above 38 C, a drop of 26.5%. **That is roughly double what temperature alone explains.** A typical silicon coefficient near -0.4% per degree across a 29 degree spread predicts about 12%. The remainder is an angle of incidence confound: a cold hour at 750 W/m² is usually a winter midday with the sun high, while a hot hour at the same horizontal irradiance is more often a summer late afternoon with the sun oblique to the panels. Attributing the full 26.5% to temperature derating would overstate the effect, so it is reported as a combined effect and separating the two is left to Assessment 3.

Feature correlations against the target over daylight hours (Figure 3) rank global horizontal radiation highest at 0.900, followed by the 24 hour and 1 hour target lags at 0.863 and 0.862. The clear sky index sits lower than expected at 0.613, so raw irradiance carries more direct signal than the normalised form despite the latter's theoretical appeal. Both are retained; the model decides.

8. Analytics and machine learning

Assessment 1 feedback recorded that the modelling plan did not identify the candidate models, baselines, horizons, error metrics, prediction interval approach or model selection rule. All six are specified here and all six are implemented.

Element	Specification
Horizons	1 hour and 24 hours ahead. Features are taken at time t and the label at $t+h$, so no feature uses a value recorded after the forecast is made and no weather forecast is assumed.
Candidate model	HistGradientBoostingRegressor. Chosen because it accepts missing values natively, which matters when the clear sky index is undefined at night by construction, and because it handles the sharp non-linearity at sunrise and sunset that a linear model cannot.
Baselines	Persistence, prediction = $y(t)$. Seasonal naive, prediction = $y(t+h-24)$, the same clock hour yesterday. Climatology, the training-set mean for that month and hour. Climatology is computed from the training slice only, never the full series.
Error metrics	MAE and RMSE in kilowatts, R2, and mean bias error to detect systematic over or under forecasting. Skill score is $1 - \text{MAE_model} / \text{MAE_persistence}$.
Prediction intervals	Quantile regression at the 10th and 90th percentiles, giving a nominal 80% interval, assessed by coverage (PICP) and mean width.
Validation	Four expanding-window walk forward folds across 2019 to 2025. Training always precedes testing in time and the data is never shuffled.
Selection rule	Lowest mean MAE across all folds, with the standard deviation across folds reported alongside. A model that wins on the mean while being unstable across folds is not selected.

One methodological point is worth recording because it changed the conclusion. **The model must be trained with the loss it is judged on.** The gradient booster defaults to squared error. Trained that way and reported on MAE, the 24 hour model produced a skill score of -0.219, worse than persistence, and an unstable fold 3 at MAE 13.95 against 7.0 to 8.8 elsewhere. Retraining with absolute error loss reversed both: skill became +0.101 and the fold standard deviation fell from 3.096 to 0.800. The apparent 2023 data problem was an artefact of the loss function, not of the extract.

Model	1 h MAE	1 h RMSE	1 h R2	24 h MAE	24 h RMSE	24 h R2
Gradient boosting (MAE loss)	3.658	9.324	0.978	6.963	18.552	0.913
Gradient boosting (squared error)	3.605	8.948	0.980	9.440	19.537	0.903
Persistence	13.904	23.241	0.866	7.742	21.213	0.887
Seasonal naive	7.744	21.212	0.887	7.742	21.213	0.887
Climatology	11.073	21.157	0.889	11.065	21.148	0.889

Table: mean across four walk forward folds. Best MAE per horizon in bold. At the 24 hour horizon the seasonal naive baseline is identical to persistence by construction, since $y(t+h-24)$ collapses to $y(t)$; they are one baseline reported under two names.

At one hour the model reaches MAE 3.61 kW and R2 0.980, a skill score of +0.74 against persistence. Persistence is a weak baseline at hourly resolution because the diurnal cycle dominates hour to hour change, which is why seasonal naive at 7.74 kW beats it; measured against that stronger baseline the model still improves MAE by 53%.

At twenty four hours the honest position is narrower. MAE 6.96 kW against 7.74 kW for the best baseline is a 10% improvement, materially smaller than at one hour. The reason is structural rather than a modelling failure: with no numerical weather prediction input, the best available information about tomorrow at noon is largely what happened today at noon, and a model can add only limited value beyond that. This is the clearest argument for the Assessment 3 work described in Section 14.

8.1 Comparison with the published benchmark

Thuseethan et al. (2025) report **R2 0.8015 and MSE 96.13** for Master Meter 1 at this site. Because this project uses the same series, a comparison is possible. It is reported with its parity conditions rather than as a headline, because the two studies do not solve the same problem.

Setting	R2	MSE
Thuseethan et al. (2025), contemporaneous mapping	0.8015	96.13
This project, 1 hour ahead forecast	0.9774	87.17
This project, 24 hour ahead forecast	0.9175	317.60
Persistence, 1 hour	0.8610	535.09
Persistence, 24 hours	0.8857	440.53

Four conditions differ and all four are stated because omitting them would make the comparison flattering and meaningless.

Condition	Thuseethan et al. (2025)	This project	Matched?
Target series	Master Meter 1, DKASC Alice Springs	Master Meter 1, DKASC Alice Springs	Yes
Problem framing	Contemporaneous mapping: irradiance at the predicted instant is an input	Forecast ahead: no value recorded at or after the forecast time is used	No
Evaluation period	2019 to 2024	Walk forward folds spanning 2019 to 2025	Partly
Sampling interval and target definition	Not established from the data alone	Hourly mean Active Power in kW	Unknown

The second row is the material one. A nowcast that is given the irradiance at the instant it is predicting is solving an easier problem than a forecast that must anticipate it. Our R2 figures exceed theirs, but that is not evidence of a better method, and claiming it would be would be the easiest part of this report to puncture. The defensible statement is narrower: **on the same series, a forecast made with no future information achieves an R2 of 0.9774 at one hour and 0.9175 at twenty four hours**, and the published nowcast provides a useful reference point for the scale of the problem rather than a competitor to beat.

Establishing full parity requires the paper's sampling interval and exact target definition, and a re-run restricted to 2019 to 2024. That is a defined Assessment 3 task.

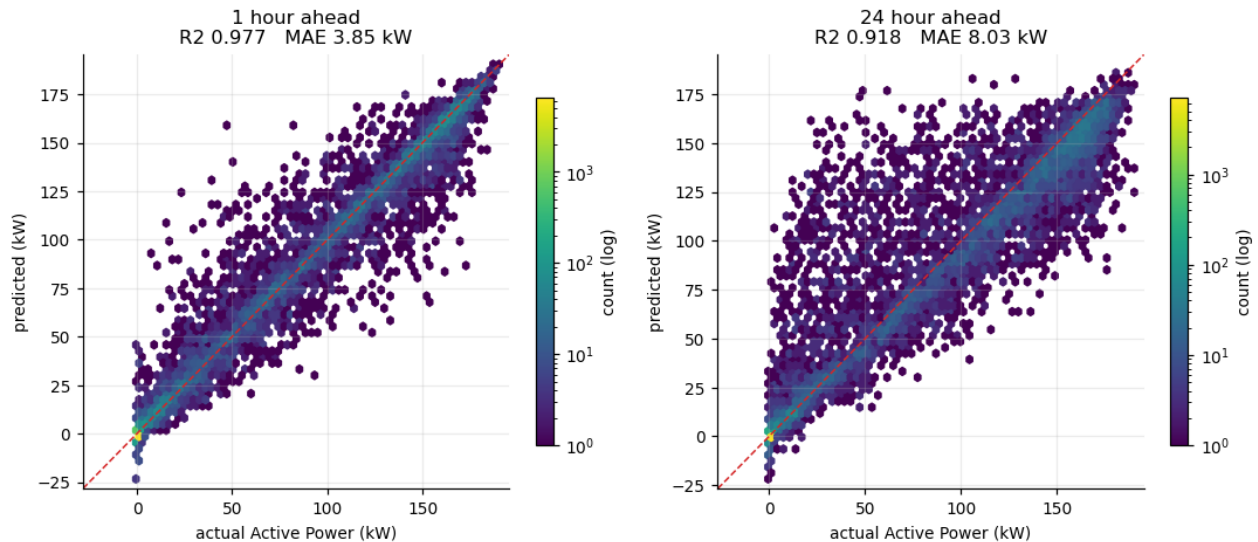


Figure 7. Predicted against actual on the final fold, log-scaled density. The 1 hour model tracks the diagonal tightly. The 24 hour model shows the characteristic fan of a forecast made without future weather, widest in the mid range where cloud decides the outcome.

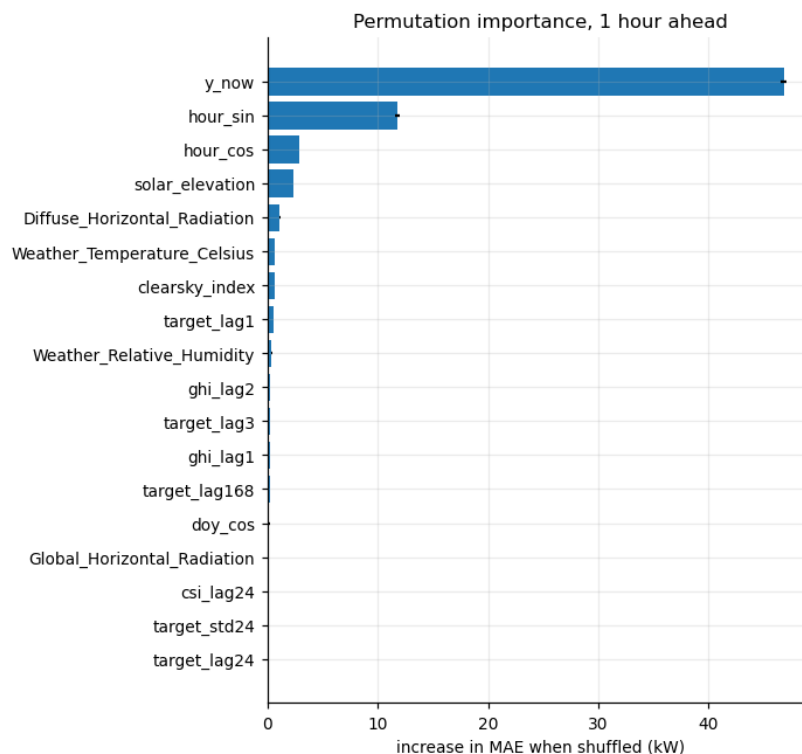


Figure 9. Permutation importance at the 1 hour horizon, measured as the increase in MAE when a feature is shuffled. Unlike split counts this reflects the effect on held-out performance.

The importance ranking is dominated by the current observation (46.83 kW increase in MAE when shuffled), followed by the hour of day encoding (11.77) and solar elevation (2.33). That ordering is correct for a one hour forecast and also a caution: at this horizon the model is largely a sophisticated persistence correction, and the

weather features earn their place mainly at the longer horizon.

9. Uncertainty, and turning a forecast into a dispatch decision

Assessment 1 feedback asked how different horizons support different decisions and how an operator should read a prediction interval when choosing to charge, hold or discharge. Both are addressed here.

Horizon	Decision it supports	How the interval is used
1 hour	Immediate charge or hold. Whether to begin absorbing an expected surplus now, or to hold charge because generation is about to fall.	The interval is narrow in daylight (mean width 24.0 kW). An operator can commit to a charging rate near the lower bound and be confident of meeting it.
24 hours	Day-ahead scheduling. Whether tomorrow warrants reserving battery headroom for a large midday surplus, or discharging tonight because tomorrow looks poor.	The interval is roughly twice as wide (mean 45.9 kW). It supports a scheduling posture, not a committed set point, and the lower bound is the planning figure.

The practical rule is that the operator plans against the **lower bound** when the cost of over-committing is high, for example when charging would otherwise be met by importing from the grid, and against the **upper bound** when the cost of under-preparing is high, for example when insufficient headroom would force curtailment of a large surplus. The point forecast is the expected case, not the planning case.

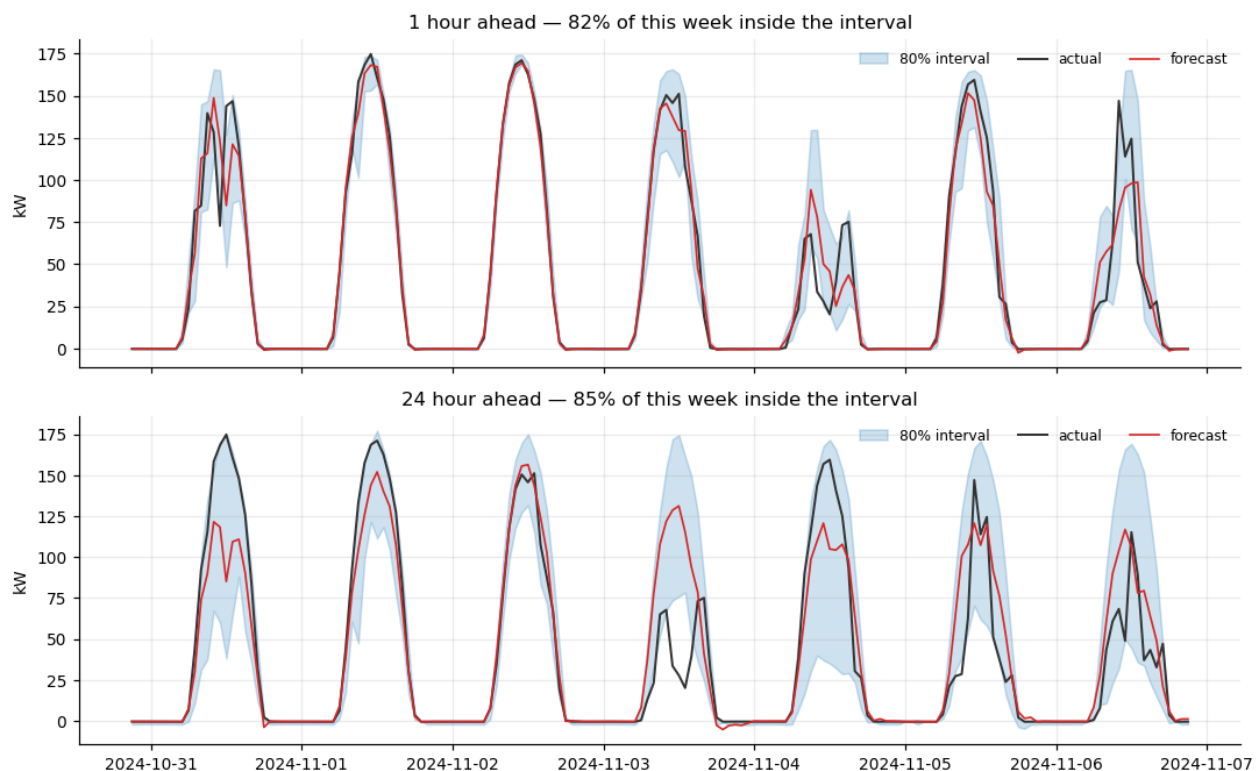


Figure 8. One week of the final fold with the 80% interval shaded. Clear days are forecast tightly at both horizons. The interval widens visibly on the unsettled days of 3 to 4 November, which is the behaviour an operator needs.

Interval quality must be reported honestly, and it is imperfect. Coverage over daylight hours is **0.772 at one hour and 0.706 at twenty four hours** against a nominal 0.80. The one hour interval is close to calibrated; the twenty four hour interval is overconfident and would mislead an operator roughly one time in ten more often than its label suggests. Widening it is straightforward and is scheduled for Assessment 3.

Coverage across all hours reads lower still, 0.702 and 0.691, but that figure is misleading and should not be quoted. At night the interval collapses to a mean width of 1.24 kW and 0.70 kW because generation is deterministically near zero, so any small wobble falls outside a nearly degenerate band. Daylight coverage is the operationally meaningful measure.

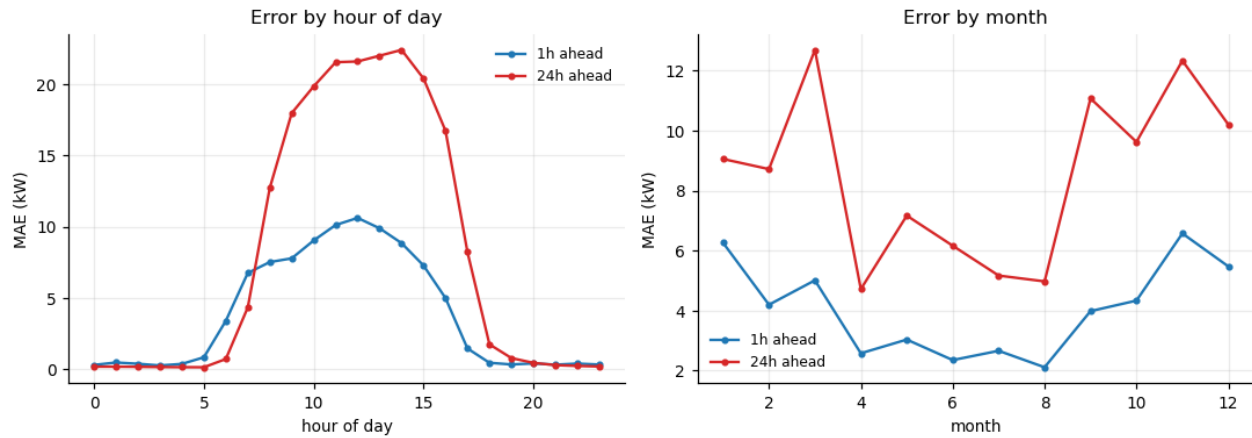


Figure 10. Mean absolute error by hour of day and by month. Error is concentrated almost entirely in daylight: 6.77 kW against 0.40 kW at night for the 1 hour model, and 14.62 kW against 0.26 kW at 24 hours.

This profile tells an operator precisely when to trust the number. Overnight the forecast is effectively exact and requires no interval at all. Between 07:00 and 17:00 the error is material and the interval carries real information. Seasonally, error peaks in the monsoon-influenced months of November through March, when convective cloud is least predictable, and is lowest in the clear dry season.

10. Implementation progress and updated workflow

The eight workflow epics named in Assessment 1 are unchanged and are used identically as Jira epics, so progress can be read directly from the board at uttamshrestha1618.atlassian.net/jira/software/projects/SCRUM/boards/1. Sprint 2, planned in August with the goal "Build the data pipeline and deliver the A2 progress report", carries thirteen tasks labelled A2 mapped to those epics.

Epic	Assessment 2 state	Evidence
Data Acquisition	Complete. All 19 extracts profiled, schema reconciliation resolved, primary dataset selected on measured grounds.	notebooks/01, 01b; SCRUM-40
Data Storage	Complete. Parquet store at two resolutions, quality baseline published.	notebooks/02; SCRUM-31, 32
Data Processing	Complete. Five stage cleaning with a measured before and after table; register repair validated independently.	notebooks/03; SCRUM-33, 34
Feature Engineering	Complete. 30 features in five families with a published dictionary and no temporal leakage.	notebooks/04; SCRUM-35
Analytics and ML	Complete for A2. Two horizons, three baselines, walk forward validation, quantile intervals, benchmark comparison.	notebooks/05; SCRUM-37, 38, 39
Visualisation	Partial. Ten static figures delivered. The interactive dashboard remains an Assessment 3 deliverable.	notebooks/04b; SCRUM-36, 22 deferred
Workflow Automation	Partial. The notebook chain is reproducible in order; scheduled orchestration is not yet implemented.	SCRUM-41, 23 carried forward
Governance and Monitoring	Complete for A2. Quality gates defined with thresholds, risk register updated, evidence links throughout.	Section 11; SCRUM-42

The pipeline is reproducible end to end from the repository. A reviewer with the raw extracts runs notebooks 01, 01b, 02, 03, 04, 04b and 05 in that order; each stage reads only the store its predecessor wrote, and each emits a markdown artefact under Outputs/ that this report cites. No figure or number here was produced outside that chain.

The honest gap is orchestration. The stages run in order because a person runs them in order, not because a scheduler enforces it. Assessment 1 committed to workflow automation as an epic and that commitment is only partly met.

11. Governance, quality gates and acceptance thresholds

Assessment 1 feedback asked for measurable acceptance thresholds, a statement of what happens when a gate fails, and a named authority for scope changes. All three follow.

Gate	Threshold	Measured now	Action on failure
Target completeness	At least 95% of hours carry a usable target over any evaluated period	99.06% PASS	Halt evaluation; re-download and re-profile before any model is retrained
Feature completeness	A feature exceeding 30% missing is excluded from the feature set and the exclusion is recorded	6 features excluded PASS	Exclusion is automatic; the feature dictionary records which and why
Register validation	Energy from the repaired register must agree with energy integrated from power within 2%	99.7% agreement PASS	Fall back to the power-integrated series and flag the register as unusable
Forecast performance	1 h MAE at or below 5.0 kW and 24 h MAE at or below 9.0 kW, with positive skill against the strongest baseline	3.66 kW and 6.96 kW PASS	Model is not promoted; the previous version remains in use and the regression is investigated
Interval coverage	Daylight PICP within 5 percentage points of nominal (0.75 to 0.85 for an 80% interval)	0.772 at 1 h PASS; 0.706 at 24 h FAIL	Interval published with an explicit warning; recalibration scheduled. Point forecasts remain usable.
Fold stability	Standard deviation of MAE across folds no greater than 20% of the mean	1 h 10.8% PASS; 24 h 11.5% PASS	Investigate the deviant fold before reporting a mean
Pipeline reproducibility	Every published figure and number regenerable from the committed notebooks	10 of 10 figures PASS	The artefact is withdrawn from the report until reproducible
Dashboard refresh	Not yet applicable; target is refresh within 60 minutes of a new extract	Deferred to A3	To be defined with the dashboard

One gate currently fails. Twenty four hour interval coverage of 0.706 sits outside the 0.75 to 0.85 band. Under the rule above the interval is published with an explicit warning rather than withdrawn, because an overconfident interval accompanied by a stated caveat is more useful to an operator than no interval at all, and the point forecast is unaffected. Recalibration is a named Assessment 3 task.

Authority. The Project Lead (Uttam Shrestha) approves changes of scope, and any such change is recorded as a dated entry under Project_Planning_Records/ with the reasoning, before the work starts. Technical decisions inside an epic rest with that epic's accountable owner. The switch of primary dataset described in Section 1 was made under this rule and is recorded accordingly. Work is complete only when it is committed, reviewed by another member through a pull request, and its Jira issue closed with an evidence link.

12. Risk

The Assessment 1 register listed nineteen risks. Four were realised during this assessment and are reported with their outcomes, and two new risks are added.

ID	Risk	Status	Outcome
T1	Schema drift, 197 columns to 2023 and 210 from 2024	Realised, worse than assessed	Actual drift is 406 distinct names with only 11 populated in every year. Mitigated by selecting a single stable extract rather than reconciling nineteen.

ID	Risk	Status	Outcome
T3	Cumulative counter misread as interval energy	Realised	The register wraps at 1,000,000 rather than counting monotonically. Differencing alone produced errors near minus one million. Repaired and validated to 99.7% against an independent estimate.
T4	High missingness: wind speed 52%, tilted radiation ~35%	Realised as forecast	Measured at 52.8% and 34% to 37%. Wind speed excluded as planned; the remainder excluded on the same documented threshold.
T5	Stale extract ending 11 February 2026	Realised, still open	Confirmed. Master Meter 1 is staler still, ending 23 August 2025. Re-download is carried into the current sprint as SCRUM-14.
T8	Overfitting through inappropriate validation	Mitigated	Four expanding-window walk forward folds, no shuffling at any point, all features backward shifted.
P1	Uneven contribution across the group	Realised	Assessment 1 attracted an individual contribution deduction for three members. Mitigated for A2 by file-level ownership, per-member Jira assignment and a documented commit allocation, so that contribution is verifiable rather than asserted.
T10	New. Training loss mismatched to the reported metric	Realised and closed	Squared error training reported on MAE produced negative skill at 24 hours and an apparent bad fold. Both resolved by matching the loss to the metric.
G5	New. Prediction intervals overconfident at long horizon	Open	24 hour daylight coverage 0.706 against nominal 0.80. Published with a warning; recalibration scheduled for A3.

13. Ethics, privacy, security and use of generative AI

The position set out in Assessment 1 is unchanged and is restated with what has since been implemented. The dataset records electrical measurements from a demonstration facility and readings from a co-located weather station. No individual is identifiable in either, no personal information is present, and no consent process is required. Data is credited to the Desert Knowledge Australia Solar Centre, and the source URL and download date are recorded so that any result can be traced to a specific extract. Raw files are not version controlled.

First Nations seasonal knowledge. The prior work on this site incorporates First Nations seasonal calendars. This project deliberately does not reproduce that approach. It is the substantive contribution of those authors, and Indigenous seasonal knowledge is cultural intellectual property that should be used only following engagement with Traditional Owners, which is beyond the scope and timeframe of this unit. The clear sky index and cyclical calendar encodings used here are derived from solar geometry and the Gregorian calendar alone.

Operational rather than statistical cost. A poor forecast has consequences beyond a worse error metric. The system is a decision support tool and not an automated controller: it emits a forecast and an interval, and a person decides. Prediction intervals are published alongside every point forecast, weak periods are stated rather than concealed (Section 9), and one governance gate is recorded as currently failing rather than quietly adjusted to pass.

Security. Credentials are never committed. The repository excludes environment files, key files, certificate files and any path matching credential patterns. This was verified during Assessment 2 when the ignore rules were amended to re-include small analysis artefacts: the credential exclusions were left intact and confirmed.

Use of generative AI. Assessment 1 committed the team to identifying any use of generative AI in accordance with CDU policy. That commitment is honoured here. Generative AI (Claude, Anthropic) was used substantially during Assessment 2 in the following ways:

Where used	Nature of the assistance	What the team did
Notebook code	Drafting the profiling, ingestion, cleaning, feature and modelling notebooks, including the plotting code.	Specified the requirements, executed every notebook, inspected all output, and identified and corrected several errors in the generated code.
Error identification	Reviewing output and flagging inconsistencies.	Four substantive errors were found and fixed this way: a sentinel rule that destroyed the energy register, an autocorrelation computed on the wrong index, a degradation trend fitted across the commissioning period, and a training loss mismatched to the reported metric. Each is documented in this report.
Report drafting	Drafting and structuring this section from the measured outputs.	Every figure and number originates from the team's own notebook runs. The team verified each against the source artefact before inclusion.
Project administration	Drafting Jira issue descriptions and the commit allocation.	Reviewed and applied by the Project Lead.

No result reported here was generated by AI without being executed and inspected by the team. Where AI-generated code proved wrong, the error and its correction are described in the body of this report rather than removed from the record.

14. Evaluation of progress, and Assessment 3

Measured against what Assessment 1 promised, the pipeline is built and reproducible, the features are implemented and documented, models are trained and validated at both stated horizons with quantified uncertainty, and the benchmark comparison is stated with its parity conditions. The dashboard and scheduled orchestration are not delivered and remain Assessment 3 work.

Four things went materially better than planned. Selecting Master Meter 1 gave the project a weather-complete dataset it would otherwise have lacked entirely. The degradation rate of -0.94% per year is a genuine result rather than a restated assumption. The register repair is validated by two independent derivations agreeing to 0.3%. And the 2024 site telemetry, discovered during profiling, makes the dispatch layer directly modellable in Assessment 3 rather than merely motivated.

Three things are weaker than they should be, and are stated rather than buried. The twenty four hour prediction interval is overconfident and one governance gate therefore fails. Workflow automation is manual. And the project's tracking discipline lapsed between assessments: Sprint 1 was left open for a month after its end date and the Assessment 2 sprint, though planned in August with the correct goal, was not started until the delivery date. The work was compressed into a single intensive sprint rather than paced across the period, which the burndown will show. The team is recording this as a process failure rather than presenting the outcome as though it were paced delivery.

The single most valuable improvement available is not a better model. It is a better input.

Assessment 3 priority	Rationale	Owner
Numerical weather prediction input at the 24 hour horizon	The 24 hour model improves on the strongest baseline by only 10% because, without a weather forecast, tomorrow's best estimate is largely today's outcome. Supplying forecast irradiance and cloud cover addresses the binding constraint directly and should yield a larger gain than any change of algorithm.	Yogesh Basnet
Recalibrate the prediction intervals	Daylight coverage of 0.706 against a nominal 0.80 fails the stated gate. Options are wider quantiles or conformal calibration on a held-out fold.	Yogesh Basnet
Build the dispatch layer on 2024 onward telemetry	BESS state of charge, site demand, PV totals and grid exchange are all measured from 2024. Surplus and headroom become directly observable, so the forecast can be translated into a charge, hold or discharge recommendation rather than left as an input to one.	Uttam Shrestha

Assessment 3 priority	Rationale	Owner
Separate temperature derating from angle of incidence	Figure 6 shows a 26.5% output difference at equal horizontal irradiance where temperature alone predicts about 12%. Computing plane-of-array irradiance would separate the two effects and likely improve the feature set.	Sarin Uprety
Refresh the extracts and automate the pipeline	Master Meter 1 ends 23 August 2025 and the yearly extracts end 11 February 2026. A scheduled rebuild would close both the staleness risk and the orchestration gap.	Abhishek Tamang
Interactive dashboard	Committed in Assessment 1 and not yet delivered. Should surface the forecast, the interval and the error profile by hour, since Figure 10 shows when the forecast can be trusted.	Yogesh Basnet

15. References

Desert Knowledge Australia Solar Centre. (n.d.). *Data download*. Retrieved 16 August 2026, from <https://dkasolarcentre.com.au/download?location=alice-springs>

Haurwitz, B. (1945). Insolation in relation to cloudiness and cloud density. *Journal of Meteorology*, 2(3), 154 to 166.

Thuseethan, S., Gangajaliya, S., Hamlin, L., Shanmugam, B., & Thennadil, S. (2025). Conv-ensemble for solar power prediction with First Nations seasonal information. *IEEE Open Journal of the Computer Society*, 6, 884 to 895. <https://doi.org/10.1109/OJCS.2025.3580339>

Repository: github.com/Uttam1618/PRT661-solar-forecasting

Jira board: uttamshrestha1618.atlassian.net/jira/software/projects/SCRUM/boards/1

Individual Work and Contribution Summary

One page per student: Laboratory 1 evidence, linked contribution, and a reflection.

Uttam Shrestha

Student ID s396560 · Project Lead · Epics: Workflow Automation, Governance and Monitoring

Laboratory 1 Completion Evidence

Laboratory 1 screenshot to be inserted here
Querying Data by Using Athena (Module 4). Must show lab name, completion status, completion score, and completion date and time.

Individual Contribution

Contribution	Evidence
Profiled all 19 raw extracts and resolved schema reconciliation, correcting the row count to a measured 1,824,033 and establishing that only 11 columns are populated in every year	notebooks/01_data_profiling.ipynb, notebooks/01b_schema_decisions.ipynb, Outputs/schema_matrix.csv, Outputs/schema_drift_summary.md
Identified that the yearly extracts carry no meteorological columns and selected Master Meter 1 as the primary dataset; recorded the change of direction	Section 1; Task_Allocation/a2-commit-allocation.md
Built the ingestion stage and the Parquet storage layer with timestamp integrity enforced against a complete 5 minute grid	notebooks/02_ingest_mastermeter1.ipynb, Outputs/data_quality_mastermeter1.md
Defined eight quality gates with quantitative thresholds, a failure action for each, and the scope-change authority rule	Section 11
Updated the risk register with four realised and two new risks; wrote the generative AI disclosure honouring the Assessment 1 commitment	Supporting_Documents/risk-register.md, ethics-privacy-security.md
Project management and report assembly: A2 sprint planning, 13 issues scoped and assigned across the eight epics, per-member commit allocation documented	Jira Sprint 2; Task_Allocation/a2-commit-allocation.md; this submission

Repository: github.com/Uttam1618/PRT661-solar-forecasting

Jira board: [SCRUM board 1](#)

Jira issues: [SCRUM-40](#), [SCRUM-41](#), [SCRUM-42](#), [SCRUM-43](#), [SCRUM-23](#)

Professional Reflection

The most useful thing I did this assessment was to profile the data before writing any pipeline code. That single decision changed the project. I expected to confirm what Assessment 1 had assumed and instead found that the archive we had planned the whole design around contains no weather data at all. Had we written the ingestion layer first and discovered that later, there would not have been time to recover.

The harder lesson concerns my own assumptions. Assessment 1 stated a row count and a schema drift figure that were both wrong, not through carelessness but because nobody had measured them. I now treat any number that cannot be traced to a committed artefact as a liability, which is why every figure here names the notebook that produced it.

On process I have to be honest: my project management was better documented than practised. This assessment's sprint was planned in August with the right goal and not started until the delivery date, and I left the previous sprint open for a month. The plan was sound; the discipline was not. For Assessment 3 I am committing to a weekly board review with the whole team present, because the governance rules I wrote are worth nothing unless someone enforces them, and that is my job.

Abhishek Tamang

Student ID S396739 · Data Engineer · Epics: Data Acquisition, Data Storage

Laboratory 1 Completion Evidence

Laboratory 1 screenshot to be inserted here
Querying Data by Using Athena (Module 4). Must show lab name, completion status, completion score, and completion date and time.

Individual Contribution

Contribution	Evidence
Executed and validated the Master Meter 1 ingestion, confirming 1,767,903 source rows reindexed onto a 1,782,422 slot five minute grid with 14,519 explicit gaps inserted	notebooks/02_ingest_mastermeter1.ipynb
Verified timestamp integrity: zero duplicate timestamps and zero unparseable timestamps across the full seventeen year period	notebooks/02, section 1 output
Produced the measured data quality baseline that every cleaning decision is justified against, including the per-column completeness table	Outputs/data_quality_mastermeter1.md
Confirmed the extract currency problem: Master Meter 1 ends 23 August 2025 and the yearly extracts end 11 February 2026, keeping risk T5 open	Section 12 of the technical report; SCRUM-14
Reviewed pull requests for the processing and modelling epics	GitHub pull request reviews

Repository: github.com/Uttam1618/PRT661-solar-forecasting

Jira board: [SCRUM board 1](#)

Jira issues: [SCRUM-31](#), [SCRUM-32](#), [SCRUM-14](#), [SCRUM-6](#)

Professional Reflection

[TO BE COMPLETED IN YOUR OWN WORDS. Suggested prompts: What surprised you about the state of the raw data? Working on acquisition and storage, what did you learn about why measuring data quality before cleaning matters? The A1 feedback found no verifiable contribution from you; what are you doing differently? What will you do differently in A3?]

One substantive observation from my epic: the value of converting implicit gaps into explicit missing values. Before reindexing onto a complete grid, a missing reading is simply an absent row and is invisible to any later stage, which means a lag feature can silently span an unknown interval. After reindexing, the 14,519 gaps are countable and every downstream decision about them is deliberate.

Sarin Uprety

Student ID S394012 · Data and Feature Engineer · Epics: Data Processing, Feature Engineering

Laboratory 1 Completion Evidence

Laboratory 1 screenshot to be inserted here
Querying Data by Using Athena (Module 4). Must show lab name, completion status, completion score, and completion date and time.

Individual Contribution

Contribution	Evidence
Executed the five stage cleaning pipeline at 5 minute resolution and produced the measured before and after table	notebooks/03_clean_mastermeter1.ipynb, Outputs/cleaning_report_mastermeter1.md
Diagnosed and repaired the cumulative energy register: a six digit counter that wraps at 1,000,000, with six wraps corrected and the result validated to 99.7% against energy integrated independently from Active Power. Closes Assessment 1 risk T3	notebooks/03, section 3 output; SCRUM-34
Established that the negative overnight register steps are real net consumption rather than corruption, preventing 43% of the record from being mislabelled	Section 5 of the technical report
Built the 30 feature hourly store across five families, with all lag and rolling features backward shifted to prevent temporal leakage	notebooks/04_features_and_ed.ipynb, Datasets/processed/features_hourly.parquet
Published the feature dictionary and confirmed the wind speed exclusion at 52.77% missing, as anticipated in Assessment 1 risk T4	Supporting_Documents/feature-dictionary.md; SCRUM-19
Recorded the year-boundary duplicate finding and the design change that resolves it	Section 4 of the technical report; SCRUM-16

Repository: github.com/Uttam1618/PRT661-solar-forecasting

Jira board: [SCRUM board 1](#)

Jira issues: [SCRUM-33](#), [SCRUM-34](#), [SCRUM-35](#), [SCRUM-19](#), [SCRUM-16](#)

Professional Reflection

[TO BE COMPLETED IN YOUR OWN WORDS. Suggested prompts: The register turned out to wrap rather than count monotonically; what did that teach you about validating an assumption before building on it? Why does cleaning order matter? The A1 feedback found no verifiable contribution from you; what are you doing differently? What will you do differently in A3?]

One substantive observation from my epic: physical bounds beat statistical ones. Setting cleaning thresholds from instrument limits and the Alice Springs climate record, rather than from the data's own percentiles, means a systematic fault cannot define what counts as normal. A value of 97,322 W/m2 in tilted radiation is not an outlier to be judged against a distribution, it is impossible, and the rule that removes it should say so.

Yogesh Basnet

Student ID S394043 · Modelling and Visualisation Lead · Epics: Analytics and Machine Learning, Visualisation

Laboratory 1 Completion Evidence

Laboratory 1 screenshot to be inserted here
Querying Data by Using Athena (Module 4). Must show lab name, completion status, completion score, and completion date and time.

Individual Contribution

Contribution	Evidence
Trained and walk forward validated gradient boosting models at both stated horizons across four expanding folds: 1 hour MAE 3.61 kW and R2 0.980, 24 hour MAE 6.96 kW and R2 0.913	notebooks/05_modelling.ipynb, Outputs/model_results.md
Implemented three baselines (persistence, seasonal naive, climatology) and established that seasonal naive, not persistence, is the strongest baseline at hourly resolution	notebooks/05, section 3 output; SCRUM-20
Identified that training loss must match the reported metric: squared error training reported on MAE gave -0.219 skill at 24 hours, corrected to +0.101 with absolute error loss, which also removed an apparent bad fold	Section 8 of the technical report; new risk T10
Produced 80% prediction intervals by quantile regression and assessed coverage by daylight and night separately, reporting the 24 hour interval as overconfident at 0.706	Outputs/model_results.md; SCRUM-38
Stated the benchmark parity conditions against Thuseethan et al. (2025) rather than reporting a favourable but non-comparable number	Section 8 of the technical report; SCRUM-21, SCRUM-39
Produced ten figures, including the corrected autocorrelation on a true hourly grid and the error profile by hour of day	notebooks/04b, 05; Outputs/figures/fig1 to fig10

Repository: github.com/Uttam1618/PRT661-solar-forecasting

Jira board: [SCRUM board 1](#)

Jira issues: [SCRUM-37](#), [SCRUM-38](#), [SCRUM-39](#), [SCRUM-36](#), [SCRUM-20](#), [SCRUM-21](#), [SCRUM-30](#)

Professional Reflection

[TO BE COMPLETED IN YOUR OWN WORDS. Suggested prompts: The 24 hour model barely beats the naive baseline; what did that teach you about where forecasting value actually comes from? Why did you report the benchmark comparison unfavourably rather than favourably? The A1 feedback found no verifiable contribution from you; what are you doing differently? What will you do differently in A3?]

One substantive observation from my epic: a model is only as good as the baseline it is measured against. Our 1 hour skill score of +0.74 looks impressive until you notice that persistence is a weak baseline at hourly resolution, because the diurnal cycle dominates hour to hour change. Against seasonal naive, the genuinely strong baseline, the improvement is 53% at one hour and only 10% at twenty four hours. Choosing the flattering baseline would have made the work look better and taught us nothing.